

Wireshark / TShark — Practical Tool Guide

Display filters, capture filters, PCAP review, field extraction, Follow Stream, statistics, and a repeatable packet-analysis workflow.

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Usage note: For analysis of provided PCAPs or traffic from a lab network you are authorized to monitor.

STATS

Understand the picture

FILTER

Isolate a protocol

STREAM

Follow a session

FIELDS

Extract evidence

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1. Capture Filter vs Display Filter

This distinction is essential.

Type	When	Example
Capture filter	Before/during capture; reduces what is captured	BPF syntax such as tcp port 80.
Display filter	During analysis after capture; does not remove data from the PCAP	Wireshark syntax such as tcp.port == 80.

Note: In TShark, -f is a capture filter and -Y is a display filter. Do not mix the syntaxes.

2. Essential display filters

Filter	Meaning
ip.addr == 10.0.0.5	Any packet where this IP is the source or destination.
ip.src == 10.0.0.5	Source IP only.
tcp.port == 80	TCP port 80 as source or destination.
udp.port == 53	Typically DNS traffic.
http.request	HTTP requests.
http.request.method == "POST"	HTTP POST requests.
dns	DNS packets.
tcp.stream eq 3	One TCP stream after identifying its stream number.

3. Read a PCAP with TShark

Useful for fast CLI review or extracting specific fields.

KALI TERMINAL — Read file

```
tshark -r capture.pcap
```

KALI TERMINAL — Display filter

```
tshark -r capture.pcap -Y "http.request"
```

KALI TERMINAL — Extract fields

```
tshark -r capture.pcap -Y "dns" -T fields -e ip.src -e dns.qry.name
```

KALI TERMINAL — Verbose field details

```
tshark -r capture.pcap -V
```

KALI TERMINAL — Protocol details

```
tshark -r capture.pcap -O http
```

4. TShark live capture

KALI TERMINAL — List interfaces

```
tshark -D
```

KALI TERMINAL — Capture and save

```
tshark -i [INTERFACE] -w out.pcap
```

KALI TERMINAL — Capture filter

```
tshark -i [INTERFACE] -f "tcp port 80" -w http.pcap
```

Note: Capture traffic only from networks/devices you are authorized to monitor.

5. Follow streams and analyze

In the GUI, Follow reconstructs a conversation so it can be read as a coherent session.

WIRESHARK — Follow TCP Stream

Right click packet → Follow → TCP Stream

WIRESHARK — Statistics

Statistics → Conversations / Endpoints / Protocol Hierarchy

Start with Statistics to see who communicates with whom, then filter, then Follow Stream - instead of reading packet by packet from the beginning.

6. Fast PCAP workflow

Step	Tool	Action
1	Protocol Hierarchy	See which protocols are present.
2	Endpoints / Conversations	Identify important IPs and ports.
3	Display Filters	Isolate HTTP, DNS, or a TCP stream.
4	Follow Stream	Read the conversation.
5	TShark fields	Extract repeated or large-value fields.
6	Document	Record packet/stream evidence.

Prepared as a quick-reference companion to the canonical Arabic guide. Use only in labs or systems you are explicitly authorized to test.

References:

<https://www.wireshark.org/docs/man-pages/wireshark-filter.html>

<https://www.wireshark.org/docs/man-pages/tshark.html>